

Human-Centered Explainable AI for Brain-Computer Interface-driven Rehabilitation

Param Rajpura
Cognitive and Brain Sciences
IIT Gandhinagar
Gandhinagar, Gujarat, India
rajpuraparam@iitgn.ac.in

Abstract

Brain-computer interfaces (BCIs) for stroke rehabilitation promise accessible, personalized therapy in resource-constrained settings. However, current systems often lack patient-facing explanations that enable users self-correction, calibrate trust, and support autonomous use. This research aims to develop human-centered explainable AI (XAI) frameworks that integrate neuroscientific validity, algorithmic interpretability, and participatory design with stroke survivors experiencing cognitive and linguistic impairments. The XAI4BCI design space is established based on completed works, and video-based methods are created to gather requirements from stroke survivors with moderate-to-severe aphasia. Initial formative co-design workshops revealed varying explainability needs among stakeholders. Ongoing work focuses on deploying adaptive XAI systems in rehabilitation settings to assess how transparency, actionability, and trust calibration influence adherence, self-efficacy, and rehabilitation outcomes. This research contributes transferable methods for inclusive AI design, theoretical frameworks for patient-facing XAI, and empirical evidence for neurotechnology deployment serving marginalized populations in the Global South healthcare contexts.

CCS Concepts

• **Human-centered computing** → **Accessibility technologies; Interaction paradigms**; • **Computing methodologies** → *Philosophical/theoretical foundations of artificial intelligence*.

Keywords

Human-centered artificial intelligence, Explainable AI, Stroke rehabilitation, Brain-computer interfaces

ACM Reference Format:

Param Rajpura. 2026. Human-Centered Explainable AI for Brain-Computer Interface-driven Rehabilitation. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3799222>

1 Introduction

Brain-computer interfaces (BCIs) for motor rehabilitation help restore limb functions through neurofeedback-driven motor imagery

training. BCIs detect neural patterns associated with the user's imagined movements and provide real-time feedback via visual displays, robotic exoskeletons, or functional electrical stimulation. This technology addresses critical rehabilitation gaps in resource-constrained settings where physiotherapist availability is severely limited. In India, for instance, there are only 0.36 physiotherapists per 10,000 people, and only 40% of stroke survivors can access rehabilitation services [2].

While BCI efficacy for motor rehabilitation has been demonstrated, current systems prioritize technical performance metrics over user experience. Explanatory feedback remains underdeveloped, and people with communication barriers are rarely involved in design processes [16]. As rehabilitation moves toward unsupervised home-based use, the absence of human-centred explanations risks system abandonment. The interpretive work that therapists perform, like diagnosing failures, prescribing corrective actions, calibrating hope, and contextualising performance within recovery trajectories, must increasingly be absorbed by the technology itself.

This research develops human-centered explainable AI (HCXAI) frameworks for BCI-driven rehabilitation that enable action rather than merely justify predictions. Unlike developer-facing interpretations or clinician-facing decision support interfaces, patient-facing explainability must serve three critical functions. First, it should enable self-correction for independent troubleshooting. Second, it must calibrate trust and motivation to sustain adherence across weeks of effortful practice. Third, it should support shared decision-making where people understand system limitations and can negotiate realistic goals with their caregivers [1].

2 Related Work and Positioning

2.1 XAI for BCIs

Existing work focuses primarily on developer-facing interpretability, using methods like GradCAM, SHAP, and LRP to validate the neurophysiological basis of predictions [11, 15]. While this work establishes that spatial, temporal, and spectral explanations can reveal which brain regions contribute to classification decisions, it does not address patient-facing explainability requirements.

2.2 Participatory Design in Healthcare

Co-design methods have successfully engaged patients in interface design but rarely address algorithmic properties like transparency and trust calibration [10]. Work with people experiencing aphasia demonstrates the necessity of communication scaffolding, but has not been applied to XAI requirements elicitation.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799222>

2.3 Human-Centred XAI

Recent CHI workshops have emphasised on Human-Centred Explainable AI (HCXAI), an approach that foregrounds human experience, context, and agency in how AI systems communicate their reasoning. The explanations must move beyond merely opening algorithmic black boxes toward supporting meaningful action and trust in the people affected by those systems [7]. However, systematic integration of HCXAI and co-design remains underdeveloped in BCI research. Few studies map stakeholder preferences on explainability or translate these into actionable design knowledge. This research bridges these gaps by positioning HCXAI as essential for advancing BCI deployment beyond laboratory benchmarks. Neuroscience reveals which brain mechanisms drive motor recovery and AI models detect these patterns with increasing accuracy. However, without grounding in human-centred design, systems can achieve technical performance while failing the people they intend to serve.

3 Research Questions

The main aim of my dissertation is to design and evaluate brain-computer interfaces for stroke rehabilitation that are grounded in human-centred AI. To achieve this, the dissertation is structured around four research directions that move from understanding to design to evaluation, spanning the intersection of Human-Computer Interaction, Neuroscience, and AI:

- **RQ1: How do stroke survivors, caregivers, and clinicians conceptualize explainability requirements for BCI-driven rehabilitation systems?**

This RQ addresses a fundamental challenge human-centred AI design where people most affected by the systems cannot articulate preferences about algorithmic transparency. They lack conceptual frameworks for explainability. As AI starts to play a fundamental role in direct delivery of care, explainability and a human-centric approach towards design become increasingly important. The research examines how participatory methods must first scaffold understanding through concrete scenarios before eliciting requirements.

- **RQ2: What explainability principles are necessary for patient-facing BCI rehabilitation systems, and how do they differ across stakeholder groups and application contexts?**

This RQ synthesises insights from neuroethics frameworks, XAI literature, and participatory design to establish principles tailored to rehabilitation contexts. While principles defined as transparency, justification, timeliness, actionability, personalisation, trust calibration, and usability provide a necessary foundation, they are largely derived from broader contexts where users have intact cognition, domain expertise, and a discrete decision to make. Rehabilitation introduces fundamentally different conditions, such as impaired perception, sustained engagement over months, family-mediated decision-making, and deep deference to clinical authority. These contextual and individual factors do not merely modulate the application of principles. They determine which

principles matter at all and reveal that some, such as transparency, can actively conflict with how trust operates in therapeutic relationships.

- **RQ3: How can XAI systems adapt to diverse cognitive abilities, communication barriers, and technology literacy levels among stroke survivors?**

This RQ examines how personalization requirements diverge across users in ways that cannot be resolved by a single explanatory design. Exploring differences in granularity preferences, mediation modes, and trust sources that emerge empirically across participants with varying severity, technology experience, and caregiver dependence. Unlike general adaptive systems, rehabilitation XAI must personalize not only the content and format of explanations but their underlying logic.

- **RQ4: What impact do human-centered XAI interventions have on adherence, self-efficacy, and functional outcomes in BCI-driven rehabilitation systems?**

This RQ evaluates deployed systems, measuring whether explanatory feedback improves rehabilitation outcomes beyond technical performance metrics. The metrics must encompass not only behavioural indicators of engagement but also neuroplastic and functional recovery markers grounded in rehabilitation science. Where the preceding questions establish what therapeutic trust requires and how to design for it, this question asks whether getting explanation right actually changes what rehabilitation technology was always supposed to change, i.e. whether people recover.

4 Contributions to Date

4.1 XAI4BCI Design Space

We established a comprehensive framework mapping explainability dimensions across BCI stakeholders, applications, and explanation types through systematic review of 84 studies [11]. This work revealed that current research prioritizes justification for developers and researchers rather than actionability for end-users. The design space articulates four key variables: Who (stakeholders ranging from end-users to regulatory agencies), Why (purposes including justification, knowledge discovery, model improvement, and control), What (application domains spanning assistive technologies to recreational uses), and How (explanation types and interfaces).

4.2 Co-design Methodology Development

We conducted formative workshops with healthy participants representing diverse technology backgrounds to develop co-design materials grounded in seven HCXAI principles: transparency, justification, timeliness, actionability, personalization, trust calibration, and usability. The seven principles were used generatively to design scenarios for codesign, shaping what participants could respond to. A three-session progressive methodology using role-play, card sorting, and wireframe design successfully elicited nuanced XAI preferences across rehabilitation, assistive technology, and gaming domains. The dialogue-labs inspired approach [8, 9] created structured yet flexible frameworks for collaborative ideation, moving participants from individual reflection to collaborative creation.

The codesign rounds helped in understanding the need for scaffolding approaches before eliciting requirements especially with non tech-savvy cohorts.

4.3 Scaffolding Methods for Stroke Survivors

To effectively co-design with stroke-survivors, I developed a video-based scaffolding approach for eliciting XAI requirements from stroke survivors with moderate-to-severe aphasia, working with three patients and three caregivers. Since the therapeutic relationship is the social context rehabilitation AI will operate within, clinician facilitation prompted the discussions. The choice was motivated by the desire to reflect deployment reality, enable aphasia-adapted communication, and treat authority effects as analytically productive data rather than methodological noise. Four facilitation approaches proved effective: analogical bridging (mapping AI states to familiar systems like mobile phone signals), projective personas (depersonalizing sensitive topics), binary forcing (reducing cognitive load through A/B choices), and extended response time (allowing 15-30 seconds for formulation).

Critically, reflexive analysis documented facilitation biases, revealing how scaffolding can inadvertently shape responses through expert override, hypothesis confirmation, and authority effects. I established practical checks for identifying systematic bias: examining directionality (do facilitator moves push toward complexity regardless of initial preference?) and contradiction (do facilitators dismiss responses conflicting with pro-AI assumptions?). This methodological contribution makes facilitator influence visible and analyzable rather than invisible and unquestioned.

Through mixed-methods research combining structured questionnaires and semi-structured interviews with stroke survivors and family caregivers, this research identifies four dimensions of "therapeutic trust", the kind of sustained, embodied, socially embedded confidence stroke survivors need to continue engaging with rehabilitation AI through plateaus and setbacks. These dimensions, namely, perceptual transparency, motivational calibration, social mediation, and institutional legitimacy, are organised using the International Classification of Functioning, Disability and Health (ICF), connecting them to a vocabulary rehabilitation clinicians already use for planning and assessment. The contribution moves beyond intelligent user interface design to argue that XAI for rehabilitation must scaffold an entire therapeutic ecosystem, making feedback perceivable through impaired sensory systems, calibrating motivation across fluctuating performance, supporting family-mediated decision-making, and anchoring algorithmic outputs in clinical institutional authority.

4.4 Model Interpretability Benchmarks

Through empirical work, this research demonstrated that performance metrics (e.g., classification accuracy for motor imagery) on limited datasets are insufficient for benchmarking prediction models or investigating the reliability of BCI predictions [13]. Explanations in the spatial, temporal, and spectral domains are needed to validate the neurophysiological basis of model decisions. We proposed metrics to quantify spatial explanations [12], standardizing benchmarks across EEG datasets and enabling systematic comparison of explanation methods.

4.5 Functional BCI Prototype

Previous works [3–6, 14] have led to the development of a motor imagery BCI system using a 16-channel wireless EEG and a 2-degree-of-freedom upper limb exoskeleton for real-time neurofeedback-driven rehabilitation. This prototype serves as the technical foundation for deploying adaptive XAI systems in clinical contexts.

5 Next Steps and Open Questions

5.1 Multi-site Requirements Validation

The scaffolding methodology is planned to expand to multiple rehabilitation centers across India, recruiting diverse stroke survivors with varied stroke severity, aphasia types, socioeconomic backgrounds, and urban versus rural contexts. This will validate the divergent XAI preferences identified in formative work (granularity needs, trust sources, mediation modes) across larger samples to establish generalizable design principles rather than context-specific findings. An open question I am still working through is what validation means for a framework contribution of this kind.

5.2 Adaptive XAI Prototypes

XAI prototypes informed by the co-design findings will be integrated into the BCI system, enabling user-configurable explainability features. The system will support three critical adaptations. First, adaptive granularity allows users to toggle between minimal summaries and detailed per-task breakdowns based on their information processing preferences. Second, multiple mediation modes enable both direct machine guidance for autonomous self-correction and family-mediated explanations for users with lower technology self-efficacy. Third, trust pluralistic interfaces accommodate divergent trust sources, supporting users who defer to human authority alongside those who prefer quantified machine feedback. An open question I am working through is who configures these features and at what point in the process. Preliminary scaffolding work suggests users cannot easily articulate preferences without support, raising the question of whether adaptive XAI requires a separate elicitation layer before use, or whether preferences should be inferred from behaviour rather than declared.

5.3 Longitudinal Deployment Studies

Evaluations will be conducted in home-based and under-resourced clinic settings with people undergoing 6 to 12 week rehabilitation programs. Impact on adherence rates, self-efficacy scores, functional motor outcomes, neurophysiological markers of recovery and technology acceptance will be measured. By comparing XAI-enabled systems against standard BCI feedback, the effects of explanatory interventions on rehabilitation outcomes can be isolated. This evaluation addresses whether human-centered XAI translates to measurable improvements in real-world deployment contexts beyond usability metrics.

5.4 Theoretical Framework Development

On a broader context, empirical findings from these studies can be transformed into transferable HCXAI principles for high-stakes healthcare AI, moving from the patient-provider dyad to enabling the patient-AI-provider triad. Also, articulating when patient-facing

explanations should prioritize enablement (supporting action) over justification (supporting understanding), how to design for episodic versus behavioral trust divergence (what people believe is correct versus what they actually do), and strategies for balancing technical capability constraints with user-articulated requirements without imposing predetermined solutions. An open question underlying this framework is whether the enablement-justification distinction can be operationalised as a design decision or is fundamentally a clinical judgment that AI systems should defer to rather than make autonomously.

5.5 Methodological Contributions

This research will formalize reflexive analysis approaches for participatory AI research with cognitively diverse populations. This includes developing practical guidelines for researchers to identify and mitigate facilitation biases while maintaining necessary scaffolding. These methods are applicable beyond rehabilitation contexts to other vulnerable populations in AI design processes, contributing to inclusive AI scholarship. Though the boundary conditions of such transferability, particularly where the therapeutic relationship itself is the object of study, remain to be established.

5.6 Dissertation Status and Long-term Goals

I started my PhD at the Human-AI Interaction (HALx) lab, Indian Institute of Technology Gandhinagar in July 2023 and am in the third year of my PhD, funded by the institute fellowship, with an expected completion in July 2028. As I move from formative qualitative work into framework development and prototyping, my dissertation is at an inflection point where the focus is shifting from foundational findings toward design and evaluation in the Global South context. My long-term goal is to develop human-centred AI systems that are legible and trustworthy to the people most affected by them, particularly populations structurally excluded from AI design processes due to cognitive, communicative, or socioeconomic barriers. I am interested in how AI deployed in high-stakes care contexts can be designed to support agency and dignity rather than merely optimise outcomes, and in building methodological foundations that make inclusive participation in AI design tractable beyond research settings.

Acknowledgments

I want to thank my supervisor, Yogesh Kumar Meena, for his exceptional support and encouragement throughout this research journey. I also extend my sincere appreciation to collaborators and all the participants of the studies for their time and cooperation.

References

- [1] Melissa J. Armstrong. 2017. Shared Decision-Making in Stroke: An Evolving Approach to Improved Patient Care. *Stroke and Vascular Neurology* 2, 2 (2017), 84–87. doi:10.1136/svn-2017-000081
- [2] Bharati Bhavna, Sahu Kirti, and Pati Sanghamitra. 2021. Rehabilitation of Stroke Patients in India: An Exploratory Study from a National-Level Survey Data 2021. *Indian Journal of Physiotherapy and Occupational Therapy* 15, 3 (June 2021), 8–18. doi:10.37506/ijpot.v15i3.16457
- [3] Anirban Chowdhury, Yogesh Kumar Meena, Haider Raza, Braj Bhushan, Ashwani Kumar Uttam, Nirmal Pandey, Adnan Ariz Hashmi, Alok Bajpai, Ashish Dutta, and Girijesh Prasad. 2018. Active Physical Practice Followed by Mental Practice Using BCI-Driven Hand Exoskeleton: A Pilot Trial for Clinical Effectiveness and Usability. *IEEE Journal of Biomedical and Health Informatics* 22, 6 (2018), 1786–1795. doi:10.1109/JBHI.2018.2863212
- [4] Anirban Chowdhury, Shyam Sunder Nishad, Yogesh Kumar Meena, Ashish Dutta, and Girijesh Prasad. 2019. Hand-Exoskeleton Assisted Progressive Neurorehabilitation Using Impedance Adaptation Based Challenge Level Adjustment Method. *IEEE Transactions on Haptics* 12, 2 (2019), 128–140. doi:10.1109/TOH.2018.2878232
- [5] Anirban Chowdhury, Haider Raza, Yogesh Kumar Meena, Ashish Dutta, and Girijesh Prasad. 2018. Online Covariate Shift Detection-Based Adaptive Brain-Computer Interface to Trigger Hand Exoskeleton Feedback for Neuro-Rehabilitation. *IEEE Transactions on Cognitive and Developmental Systems* 10, 4 (2018), 1070–1080. doi:10.1109/TCDS.2017.2787040
- [6] Anirban Chowdhury, Haider Raza, Yogesh Kumar Meena, Ashish Dutta, and Girijesh Prasad. 2019. An EEG-EMG correlation-based brain-computer interface for hand orthosis supported neuro-rehabilitation. *Journal of Neuroscience Methods* 312 (2019), 1–11. doi:10.1016/j.jneumeth.2018.11.010
- [7] Upol Ehsan, Elizabeth A Watkins, Philipp Wintersberger, Carina Manger, Sunnie S. Y. Kim, Niels Van Berkel, Andreas Riener, and Mark O Riedl. 2024. Human-Centered Explainable AI (HCXAI): Reloading Explainability in the Era of Large Language Models (LLMs). In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 477, 6 pages. doi:10.1145/3613905.3636311
- [8] Andrés Lucero, Kirsikka Vaajakallio, and Peter Dalsgaard. 2012. The dialogue-labs method: process, space and materials as structuring elements to spark dialogue in co-design events. *CoDesign* 8, 1 (2012), 1–23. doi:10.1080/15710882.2011.609888
- [9] Yogesh Kumar Meena, Krishna Seunarine, Deepak Ranjan Sahoo, Simon Robinson, Jennifer Pearson, Chi Zhang, Matt Carnie, Adam Pockett, Andrew Prescott, Suzanne K. Thomas, Harrison Ka Hin Lee, and Matt Jones. 2020. PV-Tiles: Towards Closely-Coupled Photovoltaic and Digital Materials for Useful, Beautiful and Sustainable Interactive Surfaces. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3313831.3376368
- [10] Karyn Moffatt, Joanna McGrenere, Barbara Purves, and Maria Klawe. 2004. The participatory design of a sound and image enhanced daily planner for people with aphasia. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vienna, Austria) (CHI '04). Association for Computing Machinery, New York, NY, USA, 407–414. doi:10.1145/985692.985744
- [11] Param Rajpura, Hubert Cecotti, and Yogesh Kumar Meena. 2024. Explainable artificial intelligence approaches for brain-computer interfaces: a review and design space. *Journal of Neural Engineering* 21, 4 (aug 2024), 041003. doi:10.1088/1741-2552/ad6593
- [12] Param Rajpura, Hubert Cecotti, and Yogesh Kumar Meena. 2024. Quantifying Spatial Domain Explanations in BCI using Earth Mover's Distance. In *2024 International Joint Conference on Neural Networks (IJCNN)*. IEEE, New York, NY, USA, 1–8. doi:10.1109/IJCNN60899.2024.10650619
- [13] Param Rajpura and Yogesh Kumar Meena. 2024. Towards Optimising EEG Decoding using Post-hoc Explanations and Domain Knowledge. In *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, New York, NY, USA, 1–5. doi:10.1109/EMBC53108.2024.10781846
- [14] Dheeraj Rathee, Anirban Chowdhury, Yogesh Kumar Meena, Ashish Dutta, Suzanne McDonough, and Girijesh Prasad. 2019. Brain-Machine Interface-Driven Post-Stroke Upper-Limb Functional Recovery Correlates With Beta-Band Mediated Cortical Networks. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 27, 5 (2019), 1020–1031. doi:10.1109/TNSRE.2019.2908125
- [15] Akshay Sujatha Ravindran and Jose Contreras-Vidal. 2023. An empirical comparison of deep learning explainability approaches for EEG using simulated ground truth. *Scientific Reports* 13, 1 (2023), 17709. doi:10.1038/s41598-023-43871-8
- [16] Emre Umu. 2025. Artificial Intelligence and Health Equity for People with Disabilities: An Integrated Framework for Disability-Inclusive AI Design. *IN-QUIRY: The Journal of Health Care Organization, Provision, and Financing* 62 (2025), 00469580251365472. doi:10.1177/00469580251365472